Mohamed Sameh Elkholy
Docker – Lab 1

1:

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Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\moham> docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
ea52d2000f90: Download complete
Digest: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
PS C:\Users\moham> docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

PS C:\Users\moham> docker logs 37795166ee1f4d4989c85454af2da978cdcd1fd53616ea0d357198f92009ca4a

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PS C:\Users\moham> docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED         STATUS                     PORTS     NAMES
37795166ee1f   hello-world   "/hello"   3 minutes ago   Exited (0) 3 minutes ago             jolly_herschel
PS C:\Users\moham> docker rm 37795166ee1f4d4989c85454af2da978cdcd1fd53616ea0d357198f92009ca4a
37795166ee1f4d4989c85454af2da978cdcd1fd53616ea0d357198f92009ca4a
PS C:\Users\moham> docker rmi sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
Untagged: hello-world:latest
Deleted: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
PS C:\Users\moham>
```

2:

```
Windows PowerShell

PS C:\Users\moham> docker run -it ubuntu
root@70bba8187226:/# echo docker
docker
root@70bba8187226:/# touch "hello-docker"
root@70bba8187226:/# exit
exit
PS C:\Users\moham> docker stop edb1fa5fbb32c8aea4b585c04839b790fd10007b39425eb375311b4276f1f6b8
edb1fa5fbb32c8aea4b585c04839b790fd10007b39425eb375311b4276f1f6b8
PS C:\Users\moham> docker start edb1fa5fbb32c8aea4b585c04839b790fd10007b39425eb375311b4276f1f6b8
edb1fa5fbb32c8aea4b585c04839b790fd10007b39425eb375311b4276f1f6b8
PS C:\Users\moham> docker exec -it edb1fa5fbb32c8aea4b585c04839b790fd10007b39425eb375311b4276f1f6b8 bash
root@edb1fa5fbb32:/# docker ls hello-docker
bash: docker: command not found
root@edb1fa5fbb32:/# ls hello-docker
ls: cannot access 'hello-docker': No such file or directory
root@edb1fa5fbb32:/# ls
bin  boot  dev  etc  home  lib  lib64  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
root@edb1fa5fbb32:/# exit
exit
PS C:\Users\moham> docker stop edb1fa5fbb32c8aea4b585c04839b790fd10007b39425eb375311b4276f1f6b8
edb1fa5fbb32c8aea4b585c04839b790fd10007b39425eb375311b4276f1f6b8
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edb1fa5fbb32c8aea4b585c04839b790fd10007b39425eb375311b4276f1f6b8
PS C:\Users\moham> docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
70bba81872265c0a48d1332e15e2006ff68942911c347f0210917adf328f10c5

Total reclaimed space: 12.29kB
PS C:\Users\moham> ps -a
Get-Process : A parameter cannot be found that matches parameter name 'a'.
At line:1 char:4
+ ps -a
+    ~~
    + CategoryInfo          : InvalidArgument: (:) [Get-Process], ParameterBindingException
    + FullyQualifiedErrorId : NamedParameterNotFound,Microsoft.PowerShell.Commands.GetProcessCommand

PS C:\Users\moham> docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
```

Comment:
When the container was removed, the file hello-docker was deleted too. Containers don't save files permanently. Once removed, everything inside is gone. To keep files, you need to use volumes.

3:

```
Windows PowerShell

PS C:\Users\moham> docker run -d --name app-database -e MYSQL_ROOT_PASSWORD=P4sSw0rd0! mysql:latest
Unable to find image 'mysql:latest' locally
latest: Pulling from library/mysql
4a629f1008ff: Pulling fs layer
8314ff001dec: Pulling fs layer
Digest: sha256:e5dc14f6e01c3e577e669337d2855c6d1561b30d8ef2c592e63e4e8a9a52650a
Status: Downloaded newer image for mysql:latest
5a4c6be160a6aaf13dc842e152d360e068952c64a641e26cb77d92bb74f8b5dd
PS C:\Users\moham>
```

4:

```
PS C:\Users\moham> docker run -d --name mynginx -p 80:80 nginx
6d84db9f4f3cf8a1dde8a0fc0bdbb535379566a7671f3bc6069ea675327e43d8
PS C:\Users\moham> echo "<h1>Hello from Docker!</h1>" > index.html
PS C:\Users\moham> docker cp index.html mynginx:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to mynginx:/usr/share/nginx/html/index.html
PS C:\Users\moham> del index.html
PS C:\Users\moham> curl http://localhost

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is parsed.
      RECOMMENDED ACTION:
      Use the -UseBasicParsing switch to avoid script code execution.

      Do you want to continue?

[Y] Yes  [A] Yes to All  [N] No  [L] No to All  [S] Suspend  [?] Help (default is "N"): Y


StatusCode        : 200
StatusDescription : OK
Content           : ÿþ<h1>Hello from Docker!</h1>

RawContent        : HTTP/1.1 200 OK
                    Connection: keep-alive
                    Accept-Ranges: bytes
                    Content-Length: 60
                    Content-Type: text/html
                    Date: Fri, 06 Mar 2026 13:12:19 GMT
                    ETag: "69aad27f-3c"
                    Last-Modified: Fri, 06 Mar 2026 13...
Forms             : {}
Headers           : {[Connection, keep-alive], [Accept-Ranges, bytes], [Content-Length, 60], [Content-Type, text/html]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : System.__ComObject
RawContentLength  : 60


PS C:\Users\moham> docker commit mynginx my-nginx-image
```

```
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PS C:\Users\moham> docker cp index.html mynginx:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to mynginx:/usr/share/nginx/html/index.html
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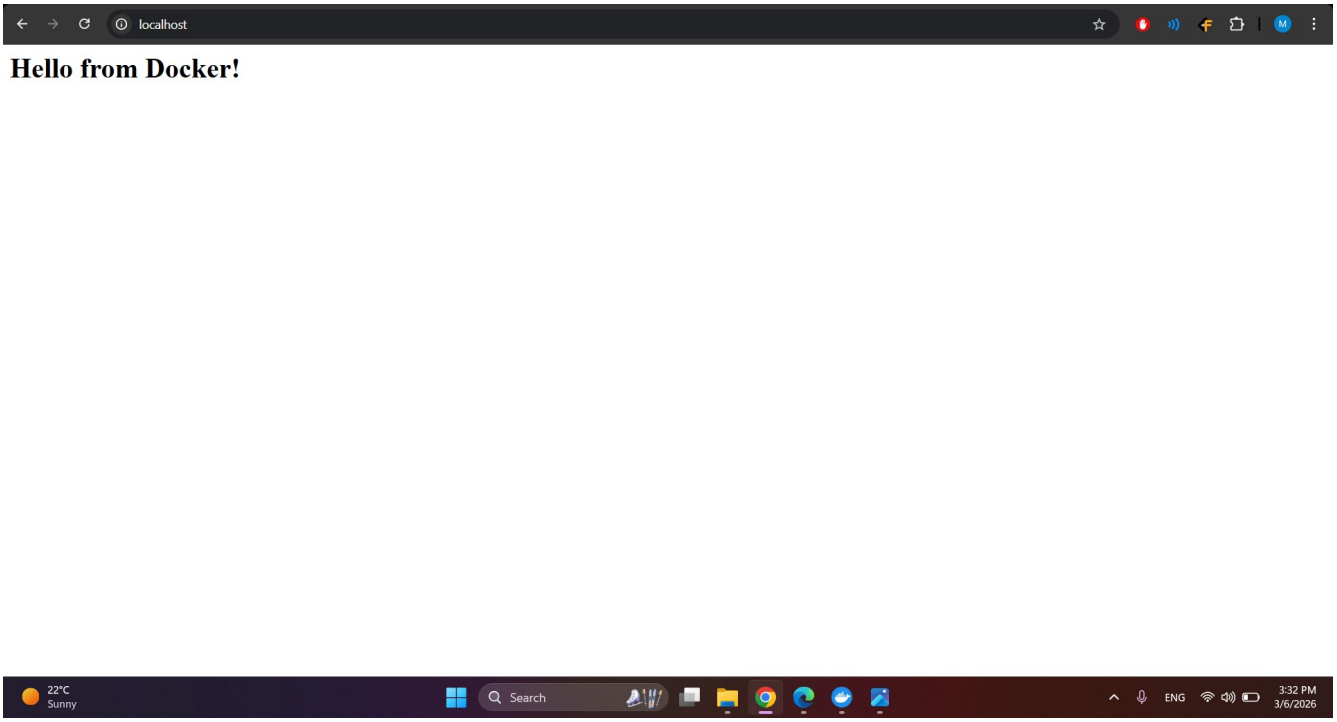
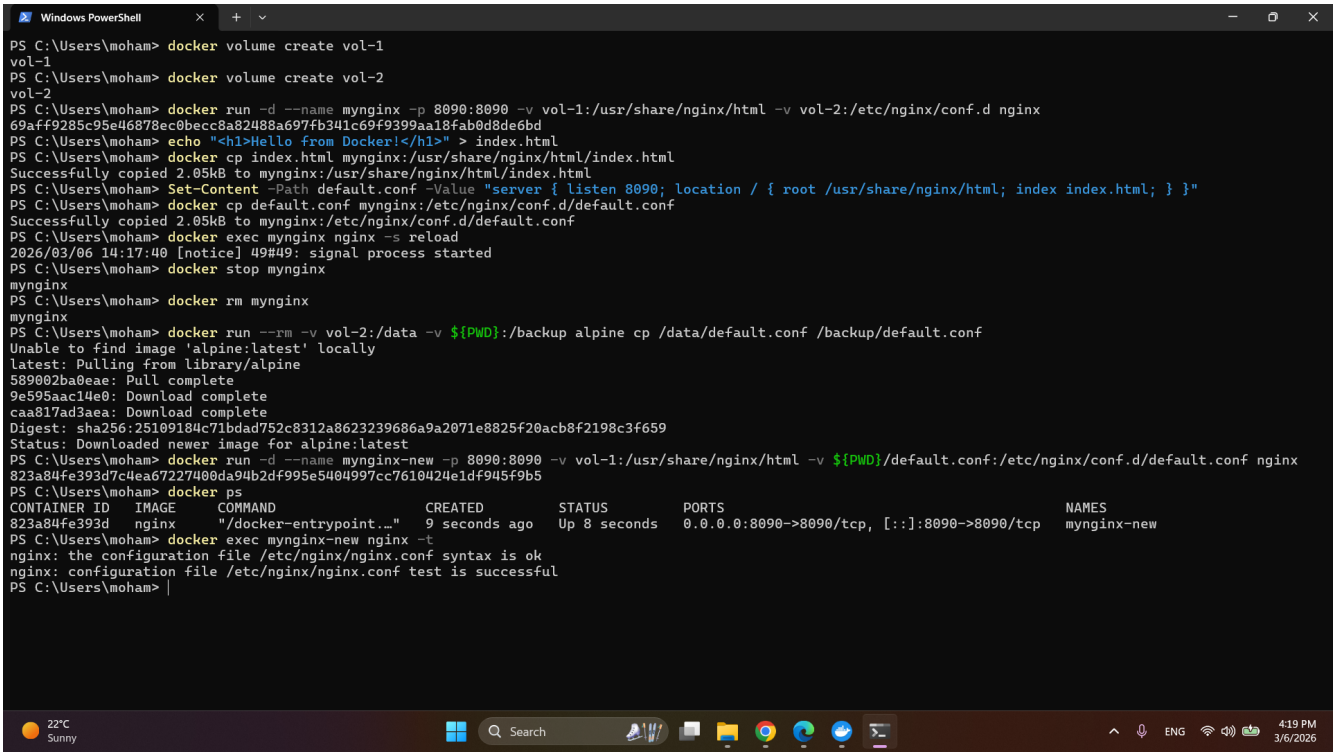
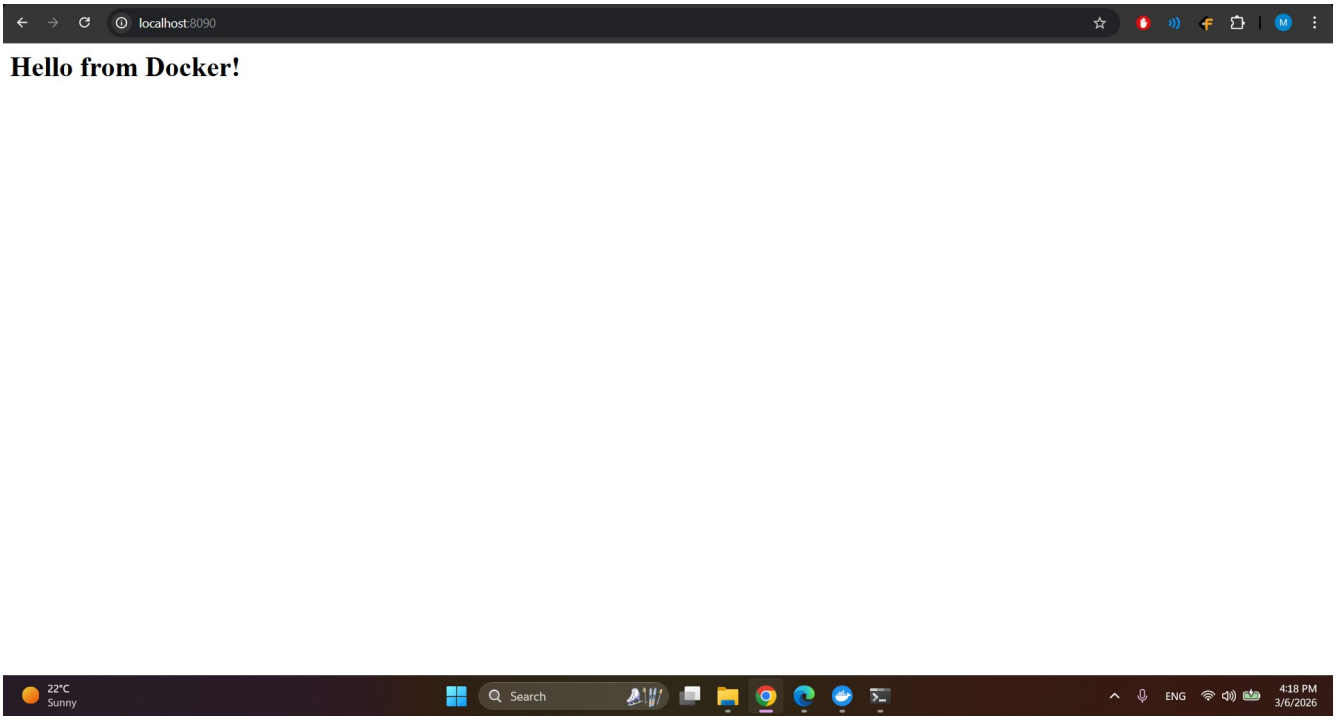
PS C:\Users\moham> docker commit mynginx my-nginx-image
sha256:ce5af3528f97de24918911d2f8e76490b7c2ab63fb00b509c327d5854ee9ae2a
PS C:\Users\moham>
```


localhost
Hello from Docker!
22°C
Sunny
Search
ENG
3:32 PM
3/6/2026

5:


Windows PowerShell
PS C:\Users\moham> docker volume create vol-1
vol-1
PS C:\Users\moham> docker volume create vol-2
vol-2
PS C:\Users\moham> docker run -d --name mynginx -p 8090:8090 -v vol-1:/usr/share/nginx/html -v vol-2:/etc/nginx/conf.d nginx
69aff9285c95e46878ec0becc8a82488a697fb341c69f9399aa18fab0d8de6bd
PS C:\Users\moham> echo "<h1>Hello from Docker!</h1>" > index.html
PS C:\Users\moham> docker cp index.html mynginx:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to mynginx:/usr/share/nginx/html/index.html
PS C:\Users\moham> Set-Content -Path default.conf -Value "server { listen 8090; location / { root /usr/share/nginx/html; index index.html; } }"
PS C:\Users\moham> docker cp default.conf mynginx:/etc/nginx/conf.d/default.conf
Successfully copied 2.05kB to mynginx:/etc/nginx/conf.d/default.conf
PS C:\Users\moham> docker exec mynginx nginx -s reload
2026/03/06 14:17:40 [notice] 49#49: signal process started
PS C:\Users\moham> docker stop mynginx
mynginx
PS C:\Users\moham> docker rm mynginx
mynginx
PS C:\Users\moham> docker run --rm -v vol-2:/data -v ${PWD}:/backup alpine cp /data/default.conf /backup/default.conf
Unable to find image 'alpine:latest' locally
latest: Pulling from library/alpine
589002ba0eae: Pull complete
9e595aac14e0: Download complete
caa817ad3aea: Download complete
Digest: sha256:25109184c71bdad752c8312a8623239686a9a2071e8825f20acb8f2198c3f659
Status: Downloaded newer image for alpine:latest
PS C:\Users\moham> docker run -d --name mynginx-new -p 8090:8090 -v vol-1:/usr/share/nginx/html -v ${PWD}/default.conf:/etc/nginx/conf.d/default.conf nginx
823a84fe393d7c4ea67227400da94b2df995e5404997cc7610424e1df945f9b5
PS C:\Users\moham> docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
823a84fe393d nginx "/docker-entrypoint.…" 9 seconds ago Up 8 seconds 0.0.0.0:8090->8090/tcp, [::]:8090->8090/tcp mynginx-new
PS C:\Users\moham> docker exec mynginx-new nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
PS C:\Users\moham>
22°C
Sunny
Search
ENG
4:19 PM
3/6/2026


localhost:8090
Hello from Docker!
22°C
Sunny
Search
ENG
4:18 PM
3/6/2026